

Zehua Jiang

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Summary

PhD candidate in Computer Science at NYU building LLM agents that act in executable environments: they call tools, read feedback from simulators and evaluators, and revise outputs until verifiable goals are met. Experience across agent frameworks, execution harnesses, SFT/GRPO fine-tuning of small LLMs, and large-scale RL in JAX.

Education

New York University, Tandon School of Engineering | Brooklyn, NY | 09/2022 - 12/2026 (expected)

Ph.D. in Computer Science and Engineering. Advisor: Julian Togelius.

Thesis: Learning and Adapting Agents for Procedural Content Generation.

New York University, Tandon School of Engineering | Brooklyn, NY | 01/2021 - 09/2022

M.S. in Electrical Engineering

Sun Yat-sen University | Guangzhou, China | 09/2016 - 06/2020

B.S. in Optoelectronic Information Science and Engineering. Exchange at UC Berkeley (01/2019 - 05/2019).

Research Experience

Ph.D. Researcher, Game Innovation Lab, New York University | Brooklyn, NY | 09/2022 - Present

Agentic PCG: tool-using LLM agents for content generation (first author, 2025 - 2026)

- Built an agent framework where an LLM observes a game level and its metrics, calls editing tools and classical generation algorithms through one interface, and an external evaluator decides which edits are kept.
- Pretrained LLMs, with no task-specific fine-tuning, met numeric targets and followed free-form language instructions across five game domains, using fewer evaluator calls than evolutionary search.
- Human study (17 participants): people picked the level generated for a given prompt in about 92% of paired comparisons.

Harness-controlled local LLM agents (first author, 2026, under review)

- Built an execution harness for local open-weight LLMs: bounded retries for malformed tool calls, keep/stop rules that prevent regressions, and extraction of numeric targets from language requests.
- With no weight updates, raised Qwen3.8-27B from 6/64 to 64/64 perfect matches on paired Zelda tasks.

Fine-tuning small LLMs for agentic generation (ongoing)

- Fine-tuning sub-4B models with SFT and GRPO on environment experience and edit traces (including traces from a trained RL agent); measuring generalization to unseen tasks, tool-call failures, latency, and memory.
- Building a library of playtested, reusable level pieces with MAP-Elites for the agent to place via tool calls.

Reinforcement learning for level generation (2022 - 2025)

- Co-developed a JAX version of PCGRL with GPU-vectorized environments (about 15x speedup) and studied what helps policies generalize to out-of-distribution map sizes and shapes (CoG 2024).
- Contributed to multi-agent RL level design with a shared policy and reward (AIIDE 2025).
- Extended PCGRL to 3D Minecraft levels with PPO agents, and evolved neural cellular automata generators with quality-diversity search (MAP-Elites, CMA-ME) (FDG 2022, first author).

Industry Experience

Research Intern, Webull Financial | St. Petersburg, FL | 06/2025 - 08/2025

- Prototyped an internal LLM agent that lets brokers query, edit, and create cash journal entries in natural language.
- Built a user-facing LLM agent that summarizes stock and options trading activity with personalized insights.
- Compared direct tool use, MCP, and LangGraph on AWS Bedrock and a Claude proxy to choose the final design.

Research Intern, Tencent Morefun Studio | Shenzhen, China | 06/2024 - 08/2024

- Worked on F.A.C.U.L., an LLM-driven FPS teammate that cooperates with players and identifies in-game objects.

Algorithm Intern, Geek+ | Beijing, China | 07/2020 - 01/2021

- Designed a multi-robot task assignment algorithm for automated warehouses that cut workstation idle time by 27.78% over a greedy baseline; validated in a live warehouse shipping 13.19 million orders.

Selected Publications

- [1] **Zehua Jiang**, Sam Earle, Ahmed Khalifa, Julian Togelius. Agentic PCG: Procedural Content Generation via Tool-using LLMs. Preprint, SSRN 6499021, 2026.
- [2] **Zehua Jiang**, Sam Earle, Ahmed Khalifa, Julian Togelius. Enabling Agentic PCG with Harness-Controlled Local Models. Preprint, SSRN 7502203, 2026.
- [3] In-Chang Baek, Sung-Hyun Kim, Sam Earle, **Zehua Jiang**, Jin-Ha Noh, Julian Togelius, Kyung-Joong Kim. PCGRLLM: Large Language Model-Driven Reward Design for Procedural Content Generation Reinforcement Learning. IEEE Transactions on Games, 2026.
- [4] Sam Earle, **Zehua Jiang**, Eugene Vinitzky, Julian Togelius. Video Game Level Design as a Multi-Agent Reinforcement Learning Problem. AIIDE 2025.
- [5] Sam Earle, **Zehua Jiang**, Julian Togelius. Scaling, Control and Generalization in Reinforcement Learning Level Generators. IEEE CoG 2024.
- [6] **Zehua Jiang**, Sam Earle, Michael Cerny Green, Julian Togelius. Learning Controllable 3D Level Generators. FDG 2022.

Invited Talk: Controllable Content Generators via Reinforcement Learning. Tutorial, IEEE CoG 2023.

Technical Skills

LLMs and Agents: tool/function calling, agent harnesses, MCP, LangGraph, SFT, GRPO, Hugging Face, AWS Bedrock

Machine Learning: PyTorch, JAX, TensorFlow; RL (PPO, multi-agent RL); quality-diversity (MAP-Elites, CMA-ME)

Programming and Tools: Python, C++, C#, JavaScript, SQL; Git, Linux, Slurm, AWS; Unity, Unreal Engine

Teaching and Service

- Student Adjunct Instructor, NYU Game Design (Spring 2025, 2026); Head TA (Fall 2024, 2025); TA (Spring 2023).
- Teaching Assistant, CS-GY 6063 Software Engineering I, NYU (Fall 2025).
- Co-organizer and Chair, Special Session on Games, IEEE WCCI 2024.
- Reviewer (50+ papers): IEEE Transactions on Games, Knowledge-Based Systems, GECCO, CoG, WCCI, AIIDE.